Moreover, Hawking was able to calculate the temperature that a far-off observer would associate with the emitted radiation and found that it is given by the strength of the gravitational field at the black hole's horizon, exactly as the analogy between the laws of black hole physics and the laws of thermodynamics suggested.3 Bekenstein was right: Hawking's results showed that the analogy should be taken seriously. In fact, these results showed that it is much more than an analogy—it is an identity. A black hole has entropy. A black hole has temperature. And the gravitational laws of black hole physics are nothing but a rewriting of the laws of thermodynamics in an extremely exotic gravitational context. This was Hawking's 1974 bombshell.

To give you a sense of the scales involved, it turns out that when one carefully accounts for all of the details, a black hole whose mass is about three times that of the sun has a temperature of about a hundred-millionth of a degree above absolute zero. It's not zero, but only just. Black holes are not black, but only barely. Unfortunately, this makes a black hole's emitted radiation meager, and impossible to detect experimentally. There is, however, an exception. Hawking's calculations also showed that the less massive a black hole is, the higher its temperature and the greater the radiation it emits. For instance, a black hole as light as a small asteroid would emit about as much radiation as a million-megaton hydrogen bomb, with radiation concentrated in the gamma-ray part of the electromagnetic spectrum. Astronomers have searched the night sky for such radiation, but except for a few long-shot possibilities they have come up empty-handed, a likely indication that such low-mass black holes, if they exist, are very rare.4 As Hawking often jokingly points out, this is too bad, for if the black hole radiation that his work predicts were to be detected, he would undoubtedly win a Nobel Prize.5

By contrast with its tiny, sub-millionth of a degree temperature, when one calculates the entropy of, say, a three-solar-mass black hole, the result is an absolutely enormous number: a one followed by about 78 zeros! And the more massive the hole, the greater the entropy. The success of Hawking's calculations unequivocally established that this truly reflects the enormous amount of disorder embodied by a black hole.